

Practical 6

17ce049

Visualizing Chipotle's Data

This time we are going to pull data directly from the internet. Special thanks to: <https://github.com/justmarkham> (<https://github.com/justmarkham>) for sharing the dataset and materials.

Step 1. Import the necessary libraries

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter

# set this so the
%matplotlib inline
```

Step 2. Import the dataset from this [address](https://raw.githubusercontent.com/justmarkham/DAT8/master/data/chipotle.tsv) (<https://raw.githubusercontent.com/justmarkham/DAT8/master/data/chipotle.tsv>).

Step 3. Assign it to a variable called chipo.

In [2]:

```
url = 'https://raw.githubusercontent.com/justmarkham/DAT8/master/data/chipotle.tsv'

chipo = pd.read_csv(url, sep = '\t')
```

Step 4. See the first 10 entries

In [134]:

```
chipo.head(10)
```

Out[3]:

	order_id	quantity	item_name	choice_description	item_price
0	1	1	Chips and Fresh Tomato Salsa	NaN	\$2.39
1	1	1	Izze	[Clementine]	\$3.39
2	1	1	Nantucket Nectar	[Apple]	\$3.39
3	1	1	Chips and Tomatillo-Green Chili Salsa	NaN	\$2.39
4	2	2	Chicken Bowl	[Tomatillo-Red Chili Salsa (Hot), [Black Beans...	\$16.98
5	3	1	Chicken Bowl	[Fresh Tomato Salsa (Mild), [Rice, Cheese, Sou...	\$10.98
6	3	1	Side of Chips	NaN	\$1.69
7	4	1	Steak Burrito	[Tomatillo Red Chili Salsa, [Fajita Vegetables...	\$11.75
8	4	1	Steak Soft Tacos	[Tomatillo Green Chili Salsa, [Pinto Beans, Ch...	\$9.25
9	5	1	Steak Burrito	[Fresh Tomato Salsa, [Rice, Black Beans, Pinto...	\$9.25

Step 5. Create a histogram of the top 5 items bought

In [4]:

```
# get the Series of the names
x = chipo.item_name

# use the Counter class from collections to create a dictionary with keys(text) and frequency
letter_counts = Counter(x)

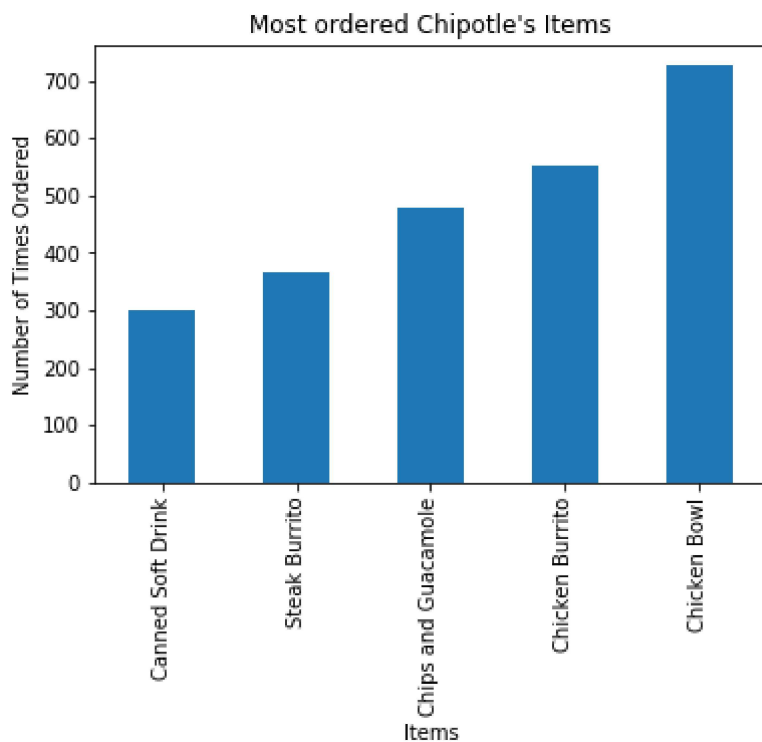
# convert the dictionary to a DataFrame
df = pd.DataFrame.from_dict(letter_counts, orient='index')

# sort the values from the top to the least value and slice the first 5 items
df = df[0].sort_values(ascending = True)[45:50]

# create the plot
df.plot(kind='bar')

# Set the title and labels
plt.xlabel('Items')
plt.ylabel('Number of Times Ordered')
plt.title('Most ordered Chipotle\'s Items')

# show the plot
plt.show()
```



Step 6. Create a scatterplot with the number of items orderered per order price

Hint: Price should be in the X-axis and Items ordered in the Y-axis

In [5]:

```
# create a list of prices
chipotle.item_price = [float(value[1:-1]) for value in chipotle.item_price] # strip the dollar sign

# then groupby the orders and sum
orders = chipotle.groupby('order_id').sum()

# creates the scatterplot
# plt.scatter(orders.quantity, orders.item_price, s = 50, c = 'green')
plt.scatter(x = orders.item_price, y = orders.quantity, s = 50, c = 'green')

# Set the title and labels
plt.xlabel('Order Price')
plt.ylabel('Items ordered')
plt.title('Number of items ordered per order price')
plt.ylim(0)
```

Out[5]:

(0, 36.7178857951459)



BONUS: Create a question and a graph to answer your own question.

In []: